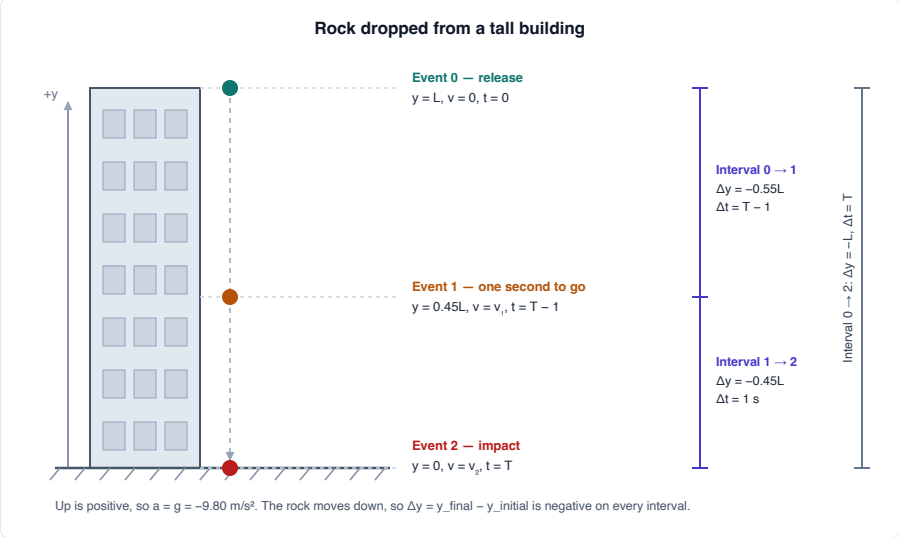
Organizing a kinematics problem - events and intervals

The following example will be used throughout:

> A rock is dropped from the top of a tall building. The rock's displacement in the last second before it hits the ground is 45% of the entire distance it falls. How tall is the building?



The idea

Unknowns live at **events** (single instants). Equations come from **intervals** (pairs of events). Separate those two things and you can tell, before doing any algebra, whether you have enough to solve the problem.

Step 1 — Event table: this is where you count unknowns

event	y	v	t
0 — release	L	0	0
1 — 1 s to go	$0.45\,L$	v_1	$T-1$
2 — impact	0	v_2	T

Each given fact is **spent filling a cell**: "dropped" $\Rightarrow v = 0$; "the last second" $\Rightarrow t = T - 1$; "45% of the fall" $\Rightarrow y = 0.45\,L$. Anything still blank gets a new symbol.

Count the new symbols: $L,\ v_1,\ v_2,\ T$ — four unknowns.

Step 2 — Interval table: this is where equations come from

Take $+y$ up, so $a = g = -9.80\ \mathrm{m/s^2}$ and $\Delta y = y_f - y_i$ is negative everywhere.

	0 → 1	1 → 2	0 → 2
Δy	$-0.55\,L$	$-0.45\,L$	$-L$
v_i	0	v_1	0
v_f	v_1	v_2	v_2
Δt	$T-1$	1 s	T

> **Two counting facts**
>
> **Each interval gives only two independent equations.** An interval has four values ($\Delta y, v_i, v_f, \Delta t$), but fixing any two determines the other two — so each interval is fully determined by only two constraints. Even through we know three formulas any two of them imply the third. Thus, writing all three equations for one interval means one of those equations gives you no new information; gains you nothing.
>
> **For this example (and similar problems)** three intervals give only four independent equations. Any column can be rebuilt from the other two ($\Delta y_{02} = \Delta y_{01} + \Delta y_{12}$, $\Delta t_{02} = \Delta t_{01} + \Delta t_{12}$, middle velocity shared). So you only need to use the equations from two columns; the third is a copy of information you already have.

Step 3 — Count, then choose

Four unknowns, four available equations, the problem is solvable. Now be strategic: **prefer equations that avoid unknowns you don't need.** Here T appears only in the Δt of intervals 0→1 and 0→2. Skip those two equations and T disappears, leaving **3 equations in 3 unknowns**.

Solving: eliminate one variable at a time

Pick a variable. Use one equation to solve for it. Substitute into **every** other equation — then throw that equation away, it's used up. If you started with n equations and n unknowns you should now have $n-1$ and $n-1$. **If the count didn't drop, you didn't eliminate anything**, you just traded one equation for an equivalent one.

Route A — intervals 0→2 and 1→2

1. $v_2^2 = 0 + 2a(-L) = 19.6\,L$
2. $v_2^2 = v_1^2 + 2a(-0.45L) = v_1^2 + 8.82\,L$
3. $v_2 = v_1 + a(1\text{ s}) = v_1 - 9.80$

(1) − (2) gives $v_1^2 = 10.78\,L$; divide by (1): $v_1^2/v_2^2 = 0.55$, so $v_1 = \sqrt{0.55}\,v_2$ (both are negative). Put that in (3): $v_2(1-\sqrt{0.55}) = -9.80$, so $v_2 = -37.9\text{ m/s}$ and $L = v_2^2/19.6 = \mathbf{73.4\text{ m}}$.

Route B — intervals 0→1 and 1→2

1. $v_1^2 = 0 + 2a(-0.55L) = 10.78\,L$
2. $v_2^2 = v_1^2 + 8.82\,L$

3. $v_2 = v_1 - 9.80$

Solve (1) for L and put it in (2): $v_2^2 = 1.818\,v_1^2$, so $v_2 = 1.348\,v_1$ (both negative). Put that in (3): $0.348\,v_1 = -9.80$, so $v_1 = -28.1\ \mathrm{m/s}$ and $L = v_1^2/10.78 = \mathbf{73.4\ m}$.

Neither route is the "right" one. They cost about the same.

Three checks

Every number in the problem must be used somewhere. If you can't point to where the 45% went, you are not finished. (A number can be "used" by filling a table cell rather than appearing in an equation, and some numbers are deliberate red herrings — but you should know which.)

Every symbol you invented must get solved for or eliminated. A v_1 you introduced and never touched again is a loose end.

Put the answer back into the original sentence. $L = 73.4$ m and $T = 3.87$ s, so the first 2.87 s covers 40.4 m and the last second covers 33.0 m. Then $33.0/73.4 = 0.45$. ✓